

LOOK in MHD Framework V3

Plug-and-play linear latent correction for missing-modality segmentation

PhD foundation: MHD V3

First side work: LOOK

Storyline for this discussion

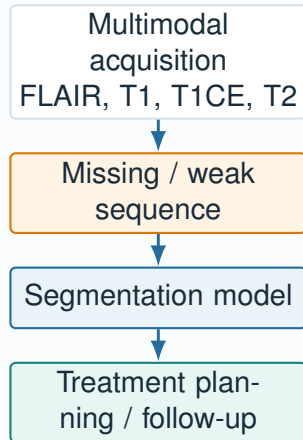
- 1 Motivation
- 2 MHD V3
- 3 LOOK
- 4 Evidence
- 5 Expansion
- 6 Discussion

One sentence

MHD V3 exposes the internal computation graph; LOOK uses that access to estimate and inject a lightweight linear correction when MRI modalities are missing.

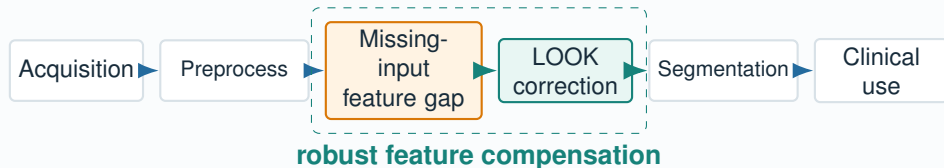
Clinical problem: multimodal MRI is informative, but often incomplete

- ▶ Complementary MRI contrasts encode different tissue and lesion characteristics.
- ▶ In practice, a sequence can be unavailable, corrupted, or omitted because of time, motion, protocol, or scanner constraints.
- ▶ A segmentation model trained for complete inputs can then receive an out-of-distribution input and lose clinically relevant information.



Take-home: The robustness problem begins at the input, but its consequences appear at the clinical endpoint.

Where LOOK sits in the medical image processing pipeline



Research positioning

LOOK is not a new segmentation backbone and not a pseudo-image generator. It estimates the missing-to-full feature discrepancy at an accessible network state and writes back a residual.

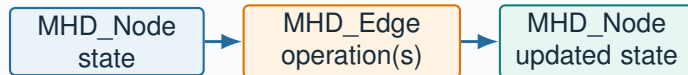
Why MHD Framework is the PhD foundation

Conventional view

A model is called as a monolithic forward function. Intermediate states are temporary and interventions require model-specific hooks or rewrites.

MHD view

A model is represented as named nodes, executable hyperedges, and level-wise topology. Intermediate states become explicit intervention points.



LOOK can read, correct, and replace a named state before selected downstream levels execute.

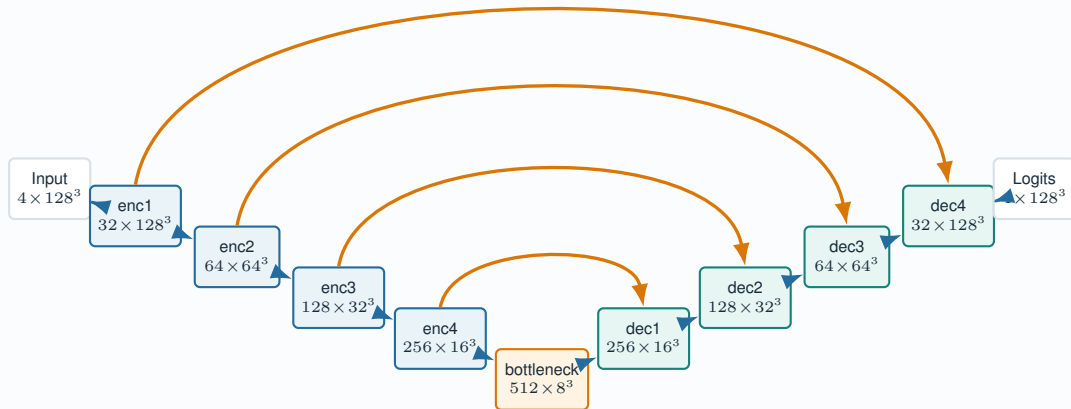
MHD V3: code-level contract used by LOOK

Component	Implemented behavior	Why it matters for LOOK
MHD_Node	<code>initial_state</code> and <code>current_state</code> ; replace/add/concat transfer modes	A stable state can be reset, observed, and overwritten with a correction.
MHD_Edge	Executes an <code>nn.Module</code> , tensor-op string, callable, or partial	Existing network operations remain reusable inside the graph.
MHD_Topo	Per-level role matrices and sort matrices encode incidence and input/output order	Multi-input decoder blocks preserve ordered skip features.
MHD_Graph	<code>forward(levels=[...])</code> runs selected execution levels	Run to a node, inject LOOK, then continue downstream.

Current graph instance

14 nodes, 12 hyperedges, and 12 execution levels, including logits, loss, and Dice states.

Current backbone: the actual 3D U-Net graph includes four skip connections



Take-home: The skip tensors are explicit second inputs to the four UpConcat hyperedges.

Problem setting in the current notebook demo

Reference path

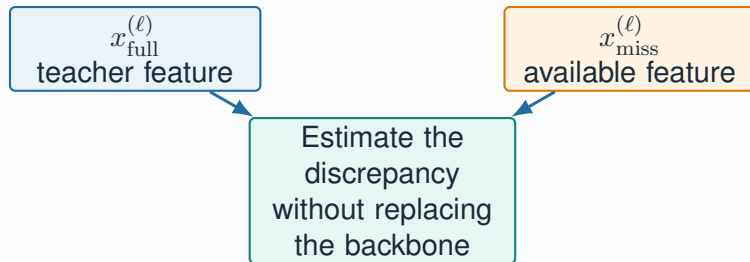
Full input: FLAIR + T1 + T1CE + T2

Feature at node ℓ : $x_{\text{full}}^{(\ell)}$

Missing path

Current pattern: FLAIR, T1, T2 removed

Only T1CE is retained: $x_{\text{miss}}^{(\ell)}$



Take-home: The present feasibility experiment studies one severe missing pattern; the method definition is broader.

LOOK: a linear residual map in a compact latent space

$$\begin{aligned}z_{\text{miss}} &= P(\mathcal{N}(D(x_{\text{miss}}))) , \\ \Delta z &= z_{\text{miss}} W + b, \\ \hat{x}_{\text{corr}} &= x_{\text{miss}} + \alpha U \left(P^{\top} \Delta z \right) .\end{aligned}$$

D/U : spatial down/up-sampling; $\mathcal{N}$: stored feature normalization; P : PCA projection; W, b : ridge correction; α : node-wise injection strength.

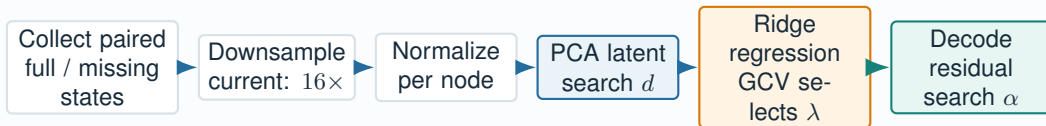
What is linear?

PCA encoding/decoding and the learned correction $zW + b$ are linear or affine. LOOK adds no nonlinear generator or new nonlinear segmentation module.

Precise claim

LOOK itself does not require nonlinear backbone retraining. The current notebook applies it to an already fine-tuned missing-pattern backbone.

How the correction is estimated

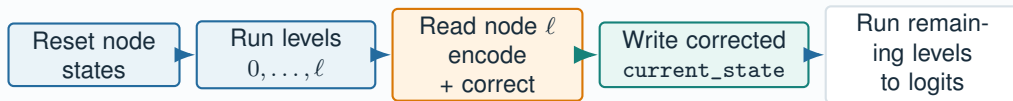


Closed-form correction

With centered latent pairs, ridge solves $W = (Z_m^\top Z_m + \lambda I)^{-1} Z_m^\top (Z_f - Z_m)$; the intercept b restores the means, and GCV selects λ .

Take-home: The expensive backbone is reused; fitting LOOK is a small paired linear estimation problem at each node.

MHD level-wise execution makes injection explicit



Candidate nodes

input, enc1–enc4, bottleneck, dec1–dec4

Key abstraction

LOOK depends on an accessible state and an execution boundary, not on a hard-coded U-Net class.

Why LOOK is designed to transfer

Plug-and-play

Fit and inject a small correction while keeping the nonlinear backbone fixed.

Composable

Apply after zero/mean filling, synthesis, modality dropout, or fine-tuning.

Inspectable

Analyze W , singular values, eigen-structure, rank, and node stability.

Boundary of the claim

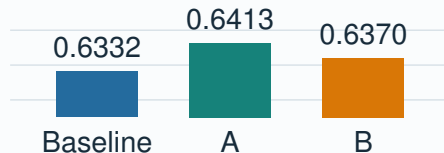
Transferability is a design hypothesis supported by a working demo. It still requires controlled experiments across missing patterns, datasets, backbones, and acquisition domains.

Current demo configuration: what has actually been run

Item	Notebook configuration
Data	BraTS 2020 training cases re-split 70/15/15: 258 train, 55 validation, 56 test
Input / task	128 ³ , four MRI channels, four-class voxel prediction; Dice reported for NCR, ED, ET
Backbone	MHD-mapped 3D U-Net; base channels 32; SGD + polynomial LR + early stopping
Missing pattern	missing_FLAIR_T1_T2: only T1CE retained
LOOK nodes	input, enc1–enc4, bottleneck, and dec1–dec4
Compression	Current spatial downsampling factor 16; PCA dimension and ridge λ searched
Status	End-to-end pipeline executed; validation improvement observed

Preliminary validation: a positive signal, not a final claim

Configuration	Mean	NCR	ED	ET
Missing baseline	0.6332	0.6213	0.5312	0.7471
LOOK, output A	0.6413	0.6249	0.5185	0.7803
LOOK, output B	0.6370	0.6246	0.5147	0.7717



Validation set only; one severe missing pattern. Output A gives $\Delta\text{Mean Dice} = +0.0081$; output B gives $+0.0038$.

Required next step

The two stored notebook outputs differ, and ED decreases while ET improves. Freeze the exact correction state, seed, and alpha dictionary, then rerun validation and untouched test evaluation.

What the current evidence does and does not establish

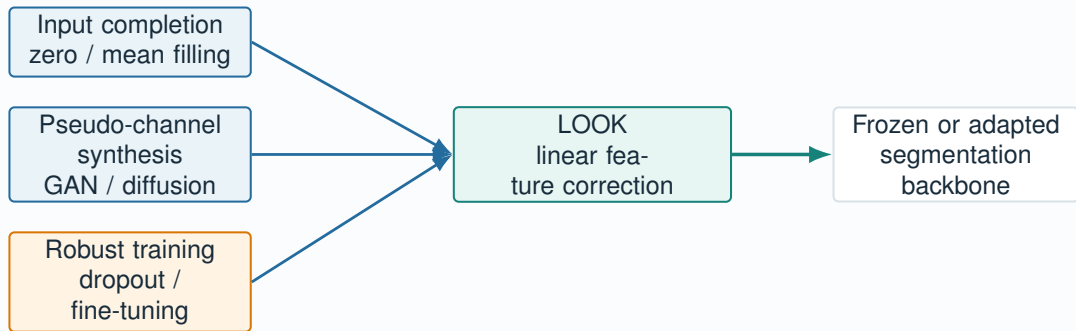
Supported now

- ▶ The full pipeline runs end-to-end.
- ▶ A linear latent residual can improve validation Mean Dice.
- ▶ Decoder-side correction can recover ET signal in this setting.
- ▶ MHD provides usable intervention points.

Not established yet

- ▶ Statistical significance or confidence intervals
- ▶ Generalization to other missing patterns
- ▶ Benefit on an untouched test set
- ▶ Superiority over synthesis or robust-training baselines
- ▶ Cross-center or cross-backbone transfer

Comparison landscape: LOOK is a correction layer, not a closed pipeline



Evaluation logic

Compare each baseline alone and with LOOK added afterward. This separates the value of input recovery or nonlinear adaptation from the incremental value of linear feature correction.

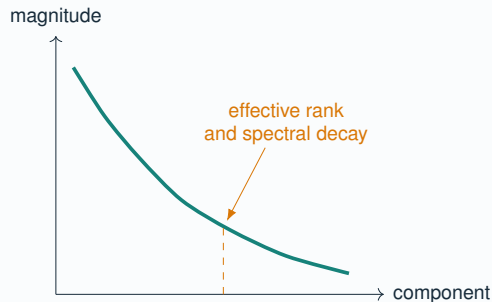
A decision-complete experiment matrix

Axis	Minimum study	Extended study	Primary readout
Missingness	Every single missing modality	Double/triple and clinically non-random patterns	Mean + class Dice; worst-case robustness
Downsampling	$2\times, 4\times, 8\times, 16\times$	$1\times$ where memory permits	Accuracy, memory, fit time, inference overhead
Injection node	Each input, encoder, bottleneck, and decoder node	Greedy and joint multi-node sets	Incremental gain and node stability
Baseline	zero, mean, modality dropout, missing-input fine-tune	GAN/diffusion and joint fine-tune	Baseline vs baseline+LOOK
Backbone	Current 3D U-Net	2D, 2.5D, large/foundation segmentation models	Transfer without nonlinear retraining
Domain	BraTS internal split	Multicenter cardiac MRI, scanner/protocol shifts	Cross-center generalization and calibration

Take-home: The paper should test a method property, not optimize a single favorable configuration.

Interpreting the correction matrix can become a scientific contribution

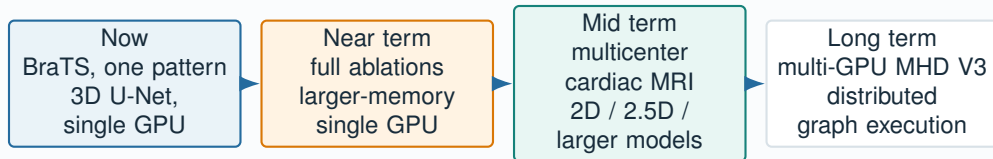
- ▶ Singular-value spectrum and effective rank of W
- ▶ Eigenvalue and eigenvector analysis for an appropriate square or symmetric operator
- ▶ Layer-wise stability across seeds, folds, centers, and missing patterns
- ▶ Principal directions associated with ET, ED, and NCR recovery
- ▶ Alignment between correction directions and full-to-missing feature shift



Interpretability question

Which low-dimensional directions consistently encode the information lost with a modality?

Roadmap: data, models, and system scale



Engineering constraint

MHD V3 is currently practical on one GPU. A larger-memory GPU is the shortest path to the next experiments; multi-GPU support should follow after the method and graph semantics are stable.

Potential TMI / MIA story: from robustness to reusable intervention

- 1 **Clinical need:** multimodal imaging pipelines must remain useful under incomplete acquisition.
- 2 **Technical gap:** many solutions retrain nonlinear generators or task networks for each setting.
- 3 **Framework contribution:** MHD exposes reusable, ordered intervention points inside arbitrary models.
- 4 **Method contribution:** LOOK estimates a compact linear residual from missing to full features.
- 5 **Evidence required:** broad missingness, strong baselines, multicenter data, multiple backbones, efficiency, uncertainty, and matrix interpretation.

Take-home: The strongest paper is not “one better Dice score”; it is a reusable and interpretable correction principle.

Decisions requested from the supervisor

Decision	Concrete question
Clinical framing	Is missing-modality segmentation the right first clinical entry point for the PhD framework?
Validation priority	First complete all BraTS missing patterns, or move earlier to multicenter cardiac data?
Method depth	Prioritize broad benchmark coverage or spectral/feature interpretation of W ?
Baseline budget	Which synthesis and robust-training baselines are essential for a credible TMI/MIA submission?
System roadmap	When should MHD V3 multi-GPU support become a research deliverable rather than infrastructure work?

Immediate next milestone

Freeze the protocol, reproduce validation, run untouched test evaluation, then expand one axis at a time.

Thank you

MHD V3 as the intervention framework
LOOK as the first linear correction study